

ALI SHAHIDI

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SUMMARY

Backend Software Engineer specializing in production AI systems, backend APIs, and distributed systems. Experienced in building FastAPI-based services, RAG pipelines, workflow orchestration systems, and real-time multimodal applications. Focused on system reliability, scalability, and AI integration in production environments.

PROFESSIONAL EXPERIENCE

Software Engineer (Research / Applied AI)

Sep 2024 – Nov 2025

ETHEREAL Research Group

Calgary, AB

- Built multimodal real-time inference pipelines (vision, audio, motion) for event detection in VR environments.
- Improved pipeline throughput by **4x** using asynchronous GPU processing and optimized frame sampling.
- Increased event detection precision by **20–25%** using cross-modal consensus and majority-voting logic.
- Designed Python backend services for real-time GPT-based dialogue and event synchronization in immersive systems.
- Reduced API cost by **30–40%** by replacing raw media transfer with structured feature extraction pipelines.
- Added structured logging and execution tracking to improve debugging and reproducibility of AI-driven workflows.

Software Engineer (Contract)

Sep 2024 – Mar 2025

ELIXR & Illumia

Calgary, AB

- Built a production RAG system using FAISS + BM25 hybrid retrieval with Reciprocal Rank Fusion (RRF).
- Improved retrieval quality using cross-encoder reranking and iterative chunk optimization (measured via Recall@K and MRR).
- Reduced LLM token usage by **40%** via selective context retrieval and ranking optimization.
- Developed FastAPI backend services for AI-powered cockpit simulation, and customer services scenario workflows.
- Implemented async API design for concurrent retrieval, inference, and response handling.

Software Engineer Intern

Feb 2024 – Aug 2024

TELUS

Calgary, AB

- Built real-time backend features supporting synchronized multi-user experiences across mobile and XR platforms.
- Implemented low-latency networking logic for distributed state synchronization.
- Contributed to backend integration and system reliability improvements for interactive applications.

Graduate Research Assistant / Software Engineer

Jan 2021 – Apr 2024

University of Calgary

Calgary, AB

- Developed software prototypes and data pipelines for assistive technology and applied systems research.
- Supported teaching in software engineering, OOP, and scalable data processing systems.

PROJECTS

AI Workflow Orchestration Middleware (Slack AI System) | FastAPI, Docker, Slack API, RBAC, LLM Tool Calling

- Built an AI-powered workflow automation system integrated with Slack for task management and execution.
- Designed LLM tool-calling pipeline with strict validation, RBAC, and policy enforcement for safe execution.
- Implemented webhook ingestion, structured logging, and audit trails for all AI-triggered actions.
- Reduced malformed AI actions using schema validation and multi-step clarification flows.

ELIXR & Illumia RAG Retrieval System | Python, FastAPI, FAISS, BM25, RRF, AWS, Docker

- Built a hybrid retrieval system combining dense (FAISS) and lexical (BM25) search.
- Built a modular FastAPI backend with async endpoints for document retrieval and ranking workflows.
- Developed a containerized backend system designed for deployment with AWS S3-based document storage.

Multimodal Behaviour Analysis System | Python, FastAPI, WhisperX, Unity, LLM, Validation

- Built real-time multimodal inference pipeline for behavior detection using vision, audio, and speech signals.

- Implemented WhisperX transcription and alignment pipeline with structured outputs.
- Designed privacy-preserving architecture with on-device feature extraction.
- Added validation and structured output constraints to ensure consistent LLM responses across multimodal inputs.

Internal AI Backend Platform | FastAPI, JWT, SQLite, OpenAI API

- Built internal backend APIs for task tracking, analytics, and AI-powered insights.
- Implemented JWT authentication, user-scoped access control, and scheduled background jobs (APScheduler).
- Developed fallback-safe AI endpoints to ensure consistent system behavior under failure conditions.

EDUCATION

University of Calgary
Master of Software Engineering (GPA: 4.0/4.0)

Apr 2024
Calgary, Canada

- Fully funded graduate program with international tuition award.
- Nominated for Distinguished Master’s Thesis Award in STEM.

University of Tehran
B.Sc. in Computer Engineering (GPA: 17.24/20)

Jul 2020
Tehran, Iran

- Awarded Best Undergraduate Project and multiple faculty fellowships.

SKILLS

Languages: Python, C#, C++, C, JavaScript, SQL
Backend: FastAPI, ASP.NET Core, REST APIs, Async Systems
AI/ML: RAG, LangChain/LangGraph, FAISS, BM25, embeddings, LLM tool calling, multimodal inference
Cloud & Infra: Docker, AWS (S3, ECS), RabbitMQ
Databases: PostgreSQL, MySQL, MS SQL, Redis
Tools: Git, Linux, Jira, Agile, LaTeX